APPLIED MATHS

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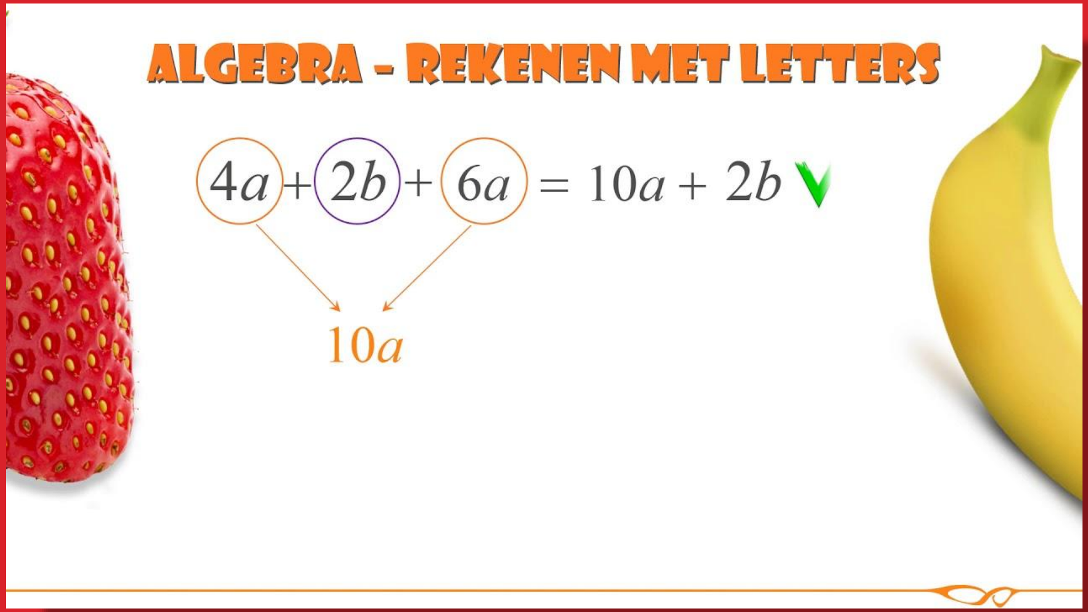
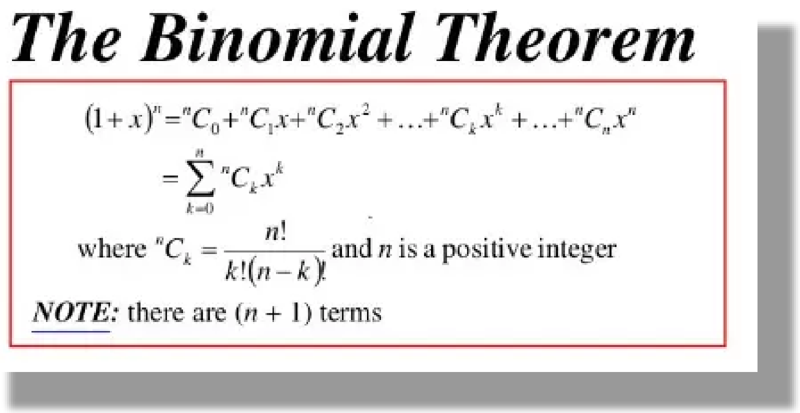
Bachelor in de elektronica-ICT Brugge – 2025-2026

HOOFDSTUK 2 : REKENEN MET LETTERVORMEN



Merkwaardige producten

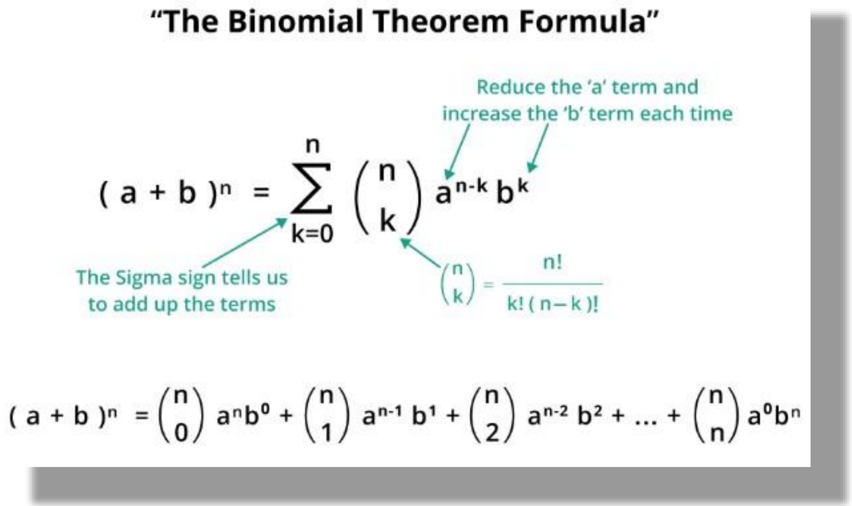
$$(a+b)^2 = a^2 + 2ab + b^2$$
$$(a-b)^2 = a^2 - 2ab + b^2$$
$$(a-b).(a+b) = a^2 - b^2$$

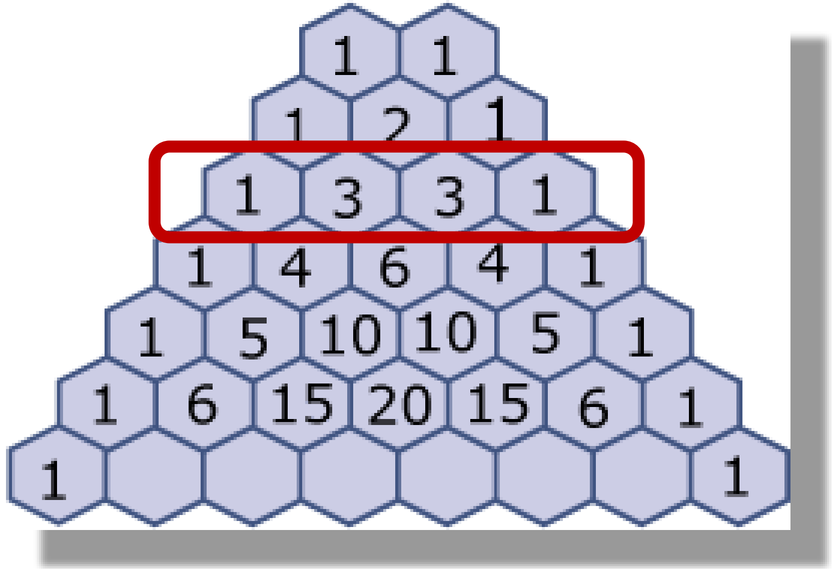


Binomium van Newton/Driehoek van Pascal

$$(a+b)^3 = 1.a^3 + 3.a^2.b + 3.a.b^2 + 1.b^3$$

$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$
$$(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$





Euclidische deling of staartdeling

$$\frac{x^3 - x + 7}{x - 1} = \dots$$

$$
\begin{array}{rl|l}
 & x^3 + 0x^2 - x + 7 & x - 1 \\
\hline
 & {\color{orange}x^3 - x^2} & {\color{orange}x^2} + {\color{blue}x} \\
- & \text{-----} & \\
 & x^2 - x & \\
 & {\color{blue}x^2 - x} & \\
- & \text{------} & \\
 & 0 + 7 &
\end{array}
$$

Euclidische deling of staartdeling

$$\frac{x^3-x+7}{x-1} = \dots$$

T(x) N(x)

$$\begin{array}{rl|l}
 & \boxed{x^3+0x^2-x+7} & \boxed{x-1} \\
\hline
 & x^3-x^2 & \boxed{x^2+x} \\
- & \text{-----} & \text{Q(x)} \\
 & x^2 \;\; -x & \\
 & x^2 \;\; -x & \\
- & \text{------} & \\
 & \boxed{0 \; + \; 7} & \\
 & \text{R(x)} &
\end{array}$$

Dus $\frac{x^2-x+7}{x-1} = x^2+x+\frac{7}{x-1}$

Stop: als graad R(x) < graad N(x)

Resultaat: $\frac{T(x)}{N(x)} = Q(x) + \frac{R(x)}{N(x)}$

Regel van Horner

$$\frac{x^3 - x + 7}{x - 1} = \ \dots$$

<u>Begin</u>

$$\begin{array}{c|ccc|c}
 & 1 & 0 & -1 & 7 \\
\hline
1 & \downarrow & & & \\
\end{array}$$

<u>Resultaat</u>

$$\begin{array}{c|ccc|c}
 & \boxed{1} & 0 & -1 & 7 \\
1 & \downarrow & 1 & 1 & 0 \\
\hline
 & \boxed{1} & 1 & 0 & 7 \\
\end{array}$$

Regel van Horner

$$\frac{x^3 - x + 7}{x - 1} = \ \ldots$$

				T(x)
N(x)	1	0	-1	7
1	↓	1	1	0
	1	1	0	7
	Q(x)			R(x)

Interpretatie

$$\frac{T(x)}{N(x)} = Q(x) + \frac{R(x)}{N(x)} \ dus \ \frac{x^3 - x + 7}{x - 1} = x^2 + x + \frac{7}{x - 1}$$

<u>Opmerking</u>: enkel bruikbaar als graad N(x) = 1

Discriminant

$$ax^2 + bx + c = a(x - x_1)(x - x_2)$$

met $D = b^2 - 4ac \qquad en \qquad x_{1,2} = \frac{-b \pm \sqrt{D}}{2a}$

Voorbeeld

$2x^2 - 10x + 12 = 2(x-2)(x-3)$

Want $D = (-10)^2 - 4.2.12 = 4$ dus $x_{1,2} = \frac{10 \pm 2}{4} = 3\ of\ 2$

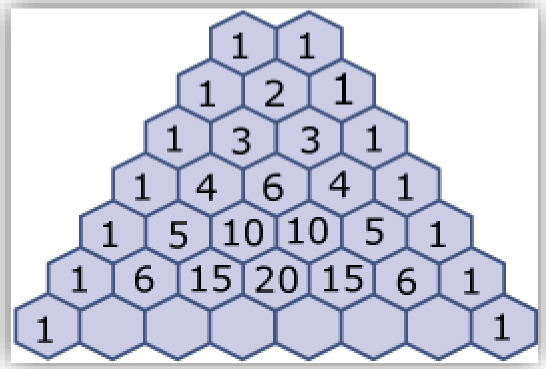
Merkwaardige producten

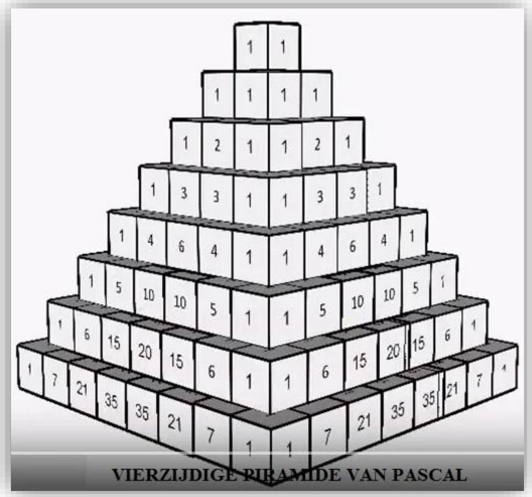
$$(a+b)^2 = a^2 + 2.ab + b^2$$

$$(a-b)^2 = a^2 - 2.ab + b^2$$

$$(a+b).(a-b) = a^2 - b^2$$

$$(x+a).(x+b) = x^2 + (a+b).x + b^2$$





$$(a+b)^3 = a^3 + 3.a^2b + 3.ab^2 + b^3$$

$$(a-b)^3 = a^3 - 3.a^2b + 3.ab^2 - b^3$$

$$(a \pm b)^3 = a^3 \pm 3.a^2b + 3.ab^2 \pm b^3$$

$$a^3 + b^3 = (a+b).(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a-b).(a^2 + ab + b^2)$$

$$(a \pm b)^4 = a^4 \pm 4.a^3b + 6.a^2b^2 \pm 4.ab^3 + b^4$$

Kwadratische veeltermen ontbinden

THE QUADRATIC FORMULA

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If $ax^2 + bx + c = 0$ but $a \neq 0$

then $\boxed{x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}}$

DISCRIMINANT

- ⇨ $b^2 - 4ac > 0$ two real solutions
- ⇨ $b^2 - 4ac = 0$ one real solutions
- ⇨ $b^2 - 4ac < 0$ zero real solutions

If the discriminant > 0,	root1 = $\dfrac{-b + \sqrt{(b - 4ac)}}{2a}$ root2 = $\dfrac{-b - \sqrt{(b - 4ac)}}{2a}$
If the discriminant = 0,	root1 = root2 = $\dfrac{-b}{2a}$
If the discriminant < 0,	root1 = $\dfrac{-b}{2a} + i\dfrac{\sqrt{-(b - 4ac)}}{2a}$ root2 = $\dfrac{-b}{2a} - i\dfrac{\sqrt{-(b - 4ac)}}{2a}$

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